Supplementary evidence

Evaluating Contextual Input and Shared Representation in Multi-Metric Runtime Forecasting for Cloud Software Systems

These post-hoc diagnostic tables accompany the original frozen experiment; they do not replace its primary estimates or prespecified classifications. All numeric inputs retain the existing result-file decimal strings in the LaTeX source; values are rounded to at most six decimal places for display. Positive gain means lower candidate MacroMAE. No additional fits, seeds, intervals or significance tests are introduced.

Original sharing comparison: matched-cell robustness

In Table S1, the candidate is M1 and reference S2. A matched cell is entity × outer fold × forecast lead, with MAE first averaged over the three existing seeds. The median uses these cell-level MAE ratios, whereas the frozen gain uses MacroMAEs. M1 win percentage counts strictly lower cell MAE; exact ties are counted separately. All 24 median ratios exceed one and all M1 win percentages are below 50%. Dependent temporal blocks, overlapping folds/leads and existing seeds are not IID replicates. These diagnostics describe direction and typical-cell magnitude, not an inferential test or a new effect-size threshold.

Table S1: Post-hoc matched-cell robustness of the original frozen M1:S2 comparison.

Block / resource	Frozen gain (%)	Median M1/S2 MAE	M1 win (%)	Ties
TimeTrack / CPU	0.469562	1.028145	35.357143	0
TimeTrack / Throughput	-0.840084	1.002157	40.357143	0
TimeTrack / Memory	-18.476886	1.192135	14.642857	0
TimeTrack / Disk write	-0.795572	1.002100	34.642857	0
fastStorage / CPU	0.045047	1.026297	32.457143	0
fastStorage / Throughput	-27.094152	1.055817	25.828571	0
fastStorage / Memory	-4.867061	1.012653	24.000000	0
fastStorage / Disk write	-4.072129	1.027516	28.914286	0
Rnd / CPU	-16.509941	1.049083	24.342857	0
Rnd / Throughput	-14.318987	1.015214	32.342857	0
Rnd / Memory	-8.915737	1.025155	23.771429	0
Rnd / Disk write	-5.551233	1.050593	18.285714	0
Materna-T1 / CPU	0.882699	1.010415	43.542857	0
Materna-T1 / Throughput	-7.308355	1.022128	22.171429	0
Materna-T1 / Memory	-13.218360	1.033259	25.257143	0
Materna-T1 / Disk write	-27.457746	1.024565	22.514286	0
Materna-T2 / CPU	-2.564363	1.020701	38.057143	0
Materna-T2 / Throughput	-8.692761	1.015360	25.942857	0
Materna-T2 / Memory	-9.266823	1.033259	25.828571	0
Materna-T2 / Disk write	-7.888503	1.014664	32.228571	0
Materna-T3 / CPU	-8.099015	1.020868	34.971429	0
Materna-T3 / Throughput	-7.914278	1.017049	26.171429	0
Materna-T3 / Memory	-8.702593	1.021046	27.657143	0
Materna-T3 / Disk write	-71.295095	1.034743	20.800000	0

Existing-seed sensitivity

Table S2 retains the existing seed-specific MacroMAE gains and their stored range and population standard deviation (SD, divisor three), expressed in percentage points (pp). Neither range nor SD is a confidence interval or standard error. FLIP denotes both strictly positive and strictly negative seed gains; "stable" means "no sign flip across the three recorded seeds", not demonstrated reproducibility under arbitrary retraining. Sign stability does not imply magnitude stability: Materna T3 M1:S2 disk-write gains are −54.65%, −71.55% and −87.84%. There are 9/24 FLIP summaries for S2:S1 and 4/24 for M1:S2. No meaningfulness threshold is imposed on small effects. Materna T1–T3 are dependent temporal blocks.

Table S2: Post-hoc sensitivity across the three existing neural seeds. Panel A: S2:S1; Panel B: M1:S2. Stable means no sign flip across the three recorded seeds; it does not imply magnitude stability.

Block / resource	Seed 0 gain (%)	Seed 1 gain (%)	Seed 2 gain (%)	Range (pp)	SD (pp)	Status
Panel A: S2:S1						
TimeTrack / CPU	-1.981093	-2.000568	-5.932358	3.951265	1.858071	stable
TimeTrack / Throughput	-0.468046	-0.188427	-0.071978	0.396068	0.166205	stable
TimeTrack / Memory	-0.785691	0.899901	2.433501	3.219192	1.314718	FLIP
TimeTrack / Disk write	-0.788174	-0.493977	-0.195247	0.592927	0.242064	stable
fastStorage / CPU	-4.653158	-0.836111	-4.163935	3.817046	1.695864	stable
fastStorage / Throughput	-2.800638	-0.007372	2.847229	5.647867	2.305777	FLIP
fastStorage / Memory	2.166251	2.660079	2.522975	0.493829	0.208144	stable
fastStorage / Disk write	2.420594	2.703501	2.940147	0.519553	0.212387	stable
Rnd / CPU	1.984487	0.394542	1.853304	1.589945	0.720580	stable
Rnd / Throughput	-5.453062	-7.570435	-8.618447	3.165385	1.316614	stable
Rnd / Memory	1.792827	1.627443	1.599031	0.193796	0.085450	stable
Rnd / Disk write	1.617878	-3.608960	2.407804	6.016764	2.669691	FLIP
Materna-T1 / CPU	-2.911522	-4.507909	-5.525094	2.613572	1.075685	stable
Materna-T1 / Throughput	1.232345	-0.027270	1.739481	1.766752	0.742760	FLIP
Materna-T1 / Memory	-0.346447	0.512781	-0.226086	0.859228	0.379866	FLIP
Materna-T1 / Disk write	-8.332861	-5.627162	-6.163909	2.705699	1.169675	stable
Materna-T2 / CPU	3.421791	4.715797	4.473312	1.294006	0.561639	stable
Materna-T2 / Throughput	1.730628	-1.933453	1.069596	3.664080	1.594461	FLIP
Materna-T2 / Memory	4.090371	3.671158	4.083098	0.419213	0.195927	stable
Materna-T2 / Disk write	-2.385167	-0.504353	-0.474830	1.910337	0.893664	stable
Materna-T3 / CPU	1.409130	-10.244289	0.420759	11.653418	5.275965	FLIP
Materna-T3 / Throughput	-1.240197	-0.483881	1.844526	3.084723	1.312716	FLIP
Materna-T3 / Memory	2.960329	2.991379	3.314328	0.353999	0.160061	stable
Materna-T3 / Disk write	-0.138669	3.366183	2.447212	3.504852	1.483812	FLIP

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Table S2 (continued)

Block / resource	Seed 0 gain (%)	Seed 1 gain (%)	Seed 2 gain (%)	Range (pp)	SD (pp)	Status
Panel B: M1:S2						
TimeTrack / CPU	1.769837	-3.222215	2.721758	5.943973	2.606776	FLIP
TimeTrack / Throughput	0.090879	-1.835451	-0.784043	1.926331	0.787521	FLIP
TimeTrack / Memory	-19.613352	-14.181809	-21.696726	7.514917	3.167826	stable
TimeTrack / Disk write	-0.322320	-1.080326	-0.985576	0.758006	0.337220	stable
fastStorage / CPU	0.590294	-0.775436	0.310718	1.365730	0.589078	FLIP
fastStorage / Throughput	-23.225501	-23.349536	-35.114890	11.889389	5.575706	stable
fastStorage / Memory	-5.076851	-4.365326	-5.158182	0.792856	0.356137	stable
fastStorage / Disk write	-4.794609	-3.879942	-3.546337	1.248271	0.527688	stable
Rnd / CPU	-16.366236	-16.420187	-16.743094	0.376858	0.166400	stable
Rnd / Throughput	-16.905837	-13.193519	-12.921484	3.984353	1.817519	stable
Rnd / Memory	-9.388542	-8.112030	-9.252328	1.276511	0.572355	stable
Rnd / Disk write	-4.995292	-5.312017	-6.368227	1.372935	0.586974	stable
Materna-T1 / CPU	-1.689360	2.006228	2.256878	3.946238	1.804100	FLIP
Materna-T1 / Throughput	-9.061155	-7.347673	-5.515523	3.545632	1.447768	stable
Materna-T1 / Memory	-13.527842	-13.443071	-12.685594	0.842248	0.378644	stable
Materna-T1 / Disk write	-28.142927	-28.070846	-26.165410	1.977516	0.915693	stable
Materna-T2 / CPU	-2.616513	-2.291012	-2.788161	0.497148	0.206174	stable
Materna-T2 / Throughput	-9.732359	-7.578477	-8.798208	2.153882	0.881891	stable
Materna-T2 / Memory	-9.337139	-8.856837	-9.608514	0.751677	0.310797	stable
Materna-T2 / Disk write	-7.648613	-8.199582	-7.820435	0.550969	0.230179	stable
Materna-T3 / CPU	-5.441695	-2.533563	-17.079941	14.546378	6.284930	stable
Materna-T3 / Throughput	-6.188420	-7.762664	-9.837298	3.648878	1.494310	stable
Materna-T3 / Memory	-6.981998	-9.316378	-9.812009	2.830011	1.233961	stable
Materna-T3 / Disk write	-54.651899	-71.546504	-87.841536	33.189637	13.550350	stable